Smart Library Request Workflow in ServiceNow

Introduction:

Managing library requests manually leads to delays, errors, and poor tracking. In many systems, students or employees request books through informal methods, and librarians handle them without a structured workflow.

This project builds an automated workflow using ServiceNow to handle book requests efficiently. It digitizes the entire process—from request submission to approval, issuing, and return—ensuring transparency, accountability, and faster operations.

Abstract:

The Smart Library Request Workflow is designed to automate and streamline library operations using ServiceNow. Users can request books through a portal, and the system automatically routes requests for approval based on predefined conditions.

The workflow includes:

- *Request submission*
- *Automated approval process*
- *Book issuance tracking*
- *Return date enforcement*
- *Status monitoring (Requested, Approved, Issued, Returned)*

By using workflow automation, the system reduces manual work, eliminates errors, and improves user experience. It also ensures that mandatory fields (like return date) are enforced using UI policies.

Objectives:

- *Eliminate manual tracking of library requests*
- *Automate approval and issuance process*
- *Ensure data accuracy using validation rules*
- *Improve response time for users*
- *Provide real-time status tracking*

Key Features:

- Automated request handling
- Role-based access (User, Librarian, Admin)
- UI Policies for mandatory fields
- Workflow automation for approvals
- Real-time tracking of request status
- Notification system (optional enhancement)

Technologies Used:

- Platform
- ServiceNow
- Modules & Concepts
- Service Catalog
- Workflow Editor / Flow Designer
- UI Policies
- Business Rules
- Tables & Fields (Custom Table: Borrow Request)

- Service Portal
- Technical Components
- Client-side scripting (UI Policies)
- Server-side logic (Business Rules)
- Database design (Tables, Relationships)

Detailed Execution of Project Phases:

Steps to Create Roles

1. Open ServiceNow
2. Click on All → search for Roles.
3. Click on All → search for Roles.
4. Select User Administration → Roles.
5. Click on New.
6. Enter Role Name as librarian → click Submit.
7. Again, click on New.
8. Enter Role Name as student → click Submit.
9. Verify that both roles (*librarian* and *student*) are listed in the Roles module.

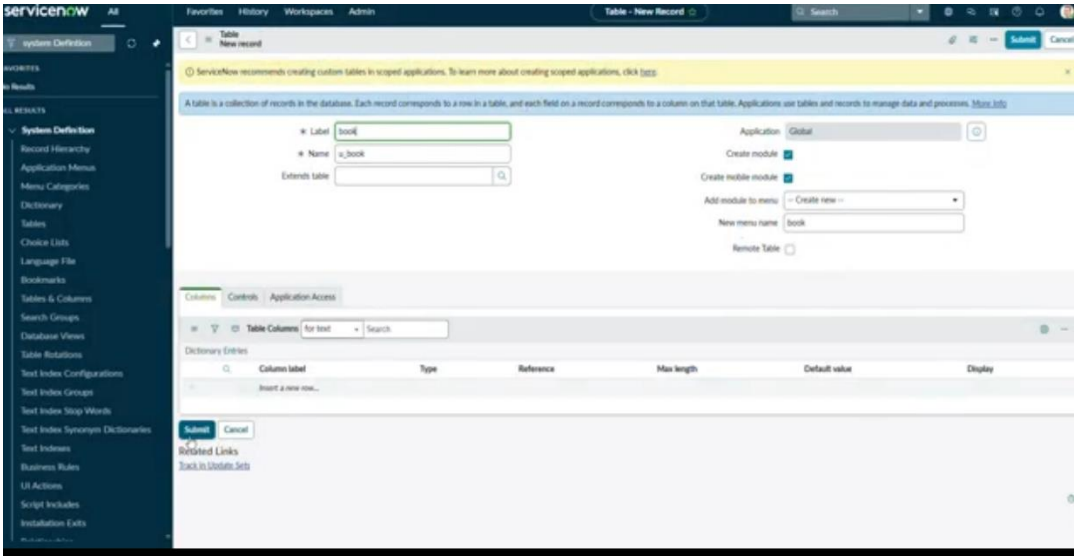
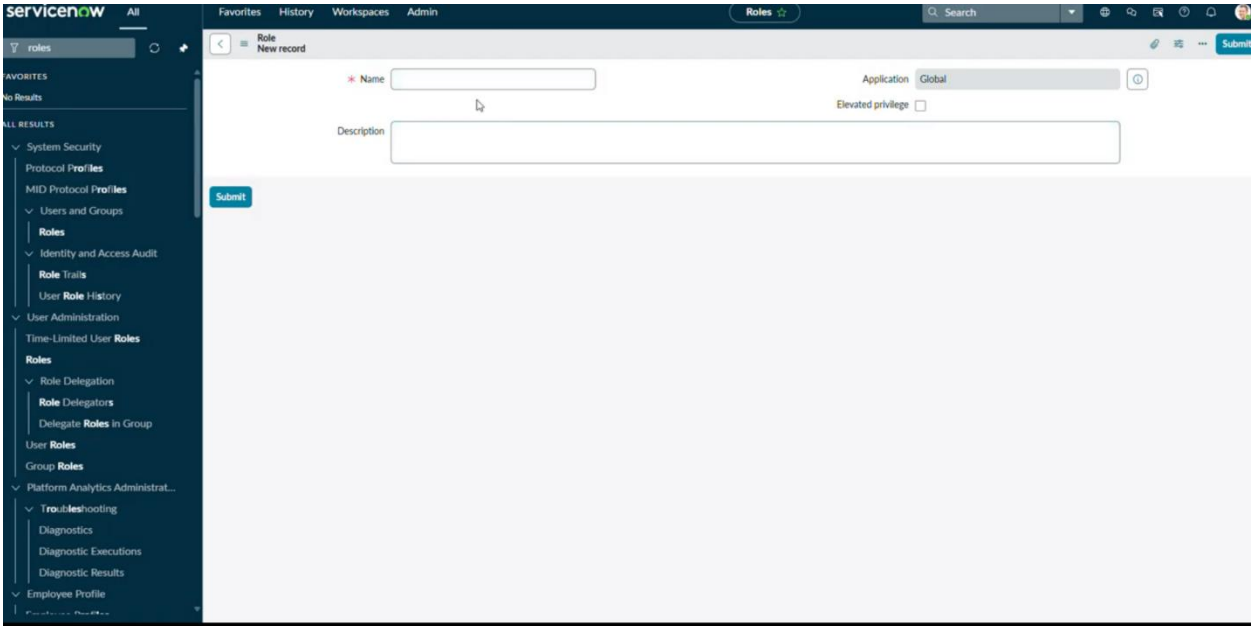


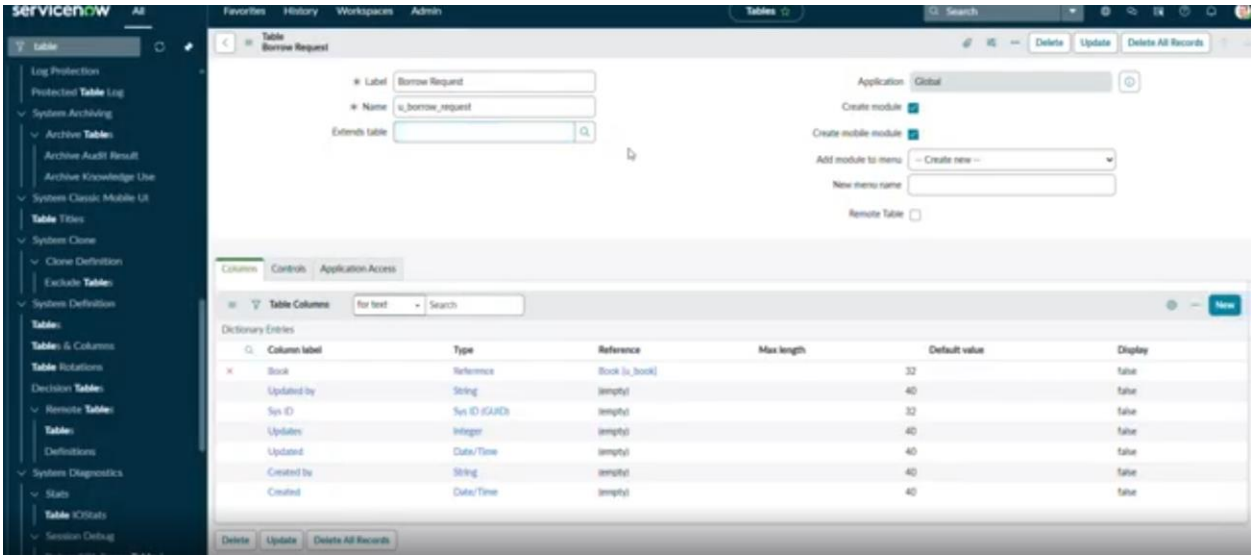
Table Creation:

1. Open ServiceNow.
2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. Click on New.
5. Enter Label as Book → click Submit (this creates the table u_book).
6. Again, click on New.
7. Enter Label as Borrow Request → click Submit (this creates the table u_borrow_request).
8. Verify that both tables (Book and Borrow Request) are listed under the Tables module.



Steps to Configure Reference Qualifier (Prevent Requesting Already-Issued Books):

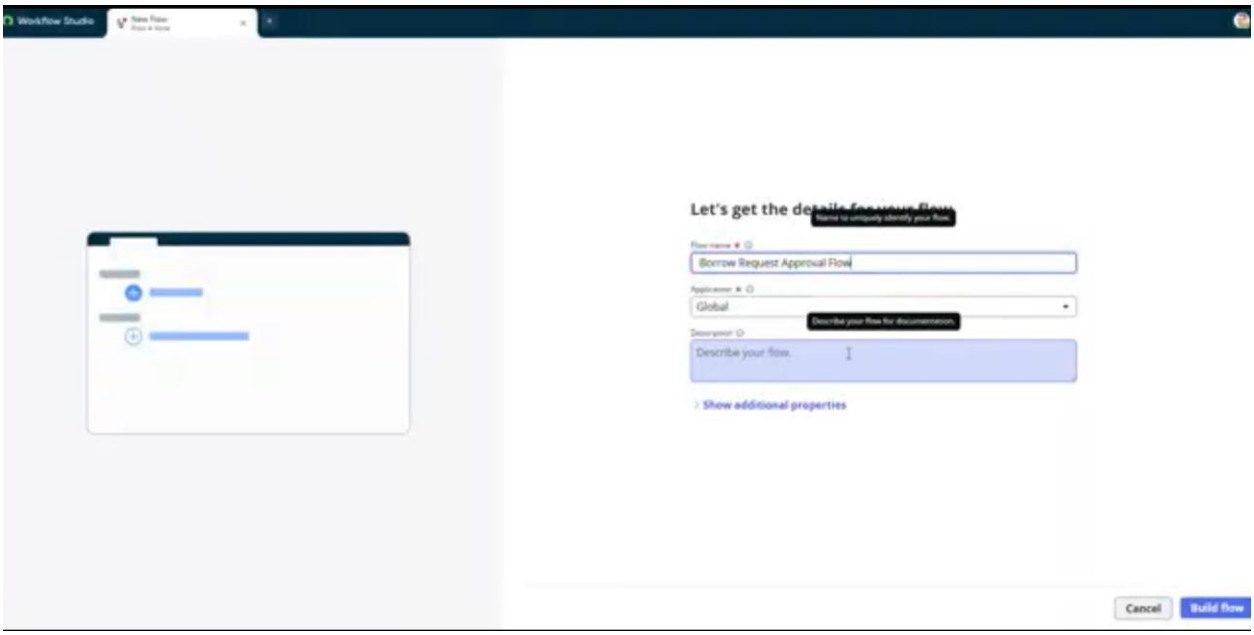
1. Open ServiceNow and click on All in the Application Navigator.
2. Search for Tables and select System Definition → Tables.
3. From the table list, open Borrow Request [u_borrow_request].
4. Scroll down to the Columns (Dictionary) related list.
5. Locate and open the field Book (Reference field to Book [u_book]).
6. In the field form, switch to Advanced View (click the gear ⚙ icon if Advanced View is not visible).
7. Scroll to the Reference Qualifier section.
8. Select Simple as the Reference Qualifier type.
9. In the Reference Qualifier Condition, set the condition as: Status is Available.
10. Click Update to save the changes.



Steps to Create Flow in Flow Designer (Approve → Issue Book):

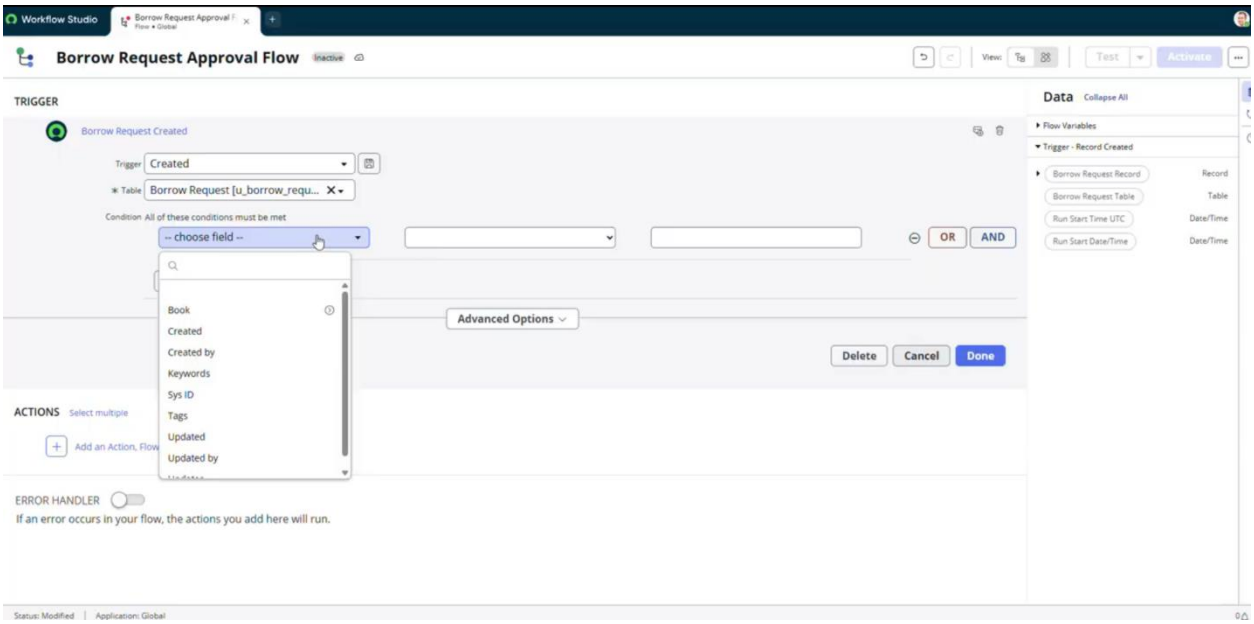
1. Open Flow Designer:

- Open ServiceNow.
- Click on All → search for Flow Designer.
- Select Flow Designer and click New → Flow.
- Enter Name as Borrow Request Approval Flow → click Submit.



Add Trigger (When to Run the Flow):

- In the Flow Designer canvas, click + Add Trigger.
- Select Created or Updated.
- Choose Table = Borrow Request [u_borrow_request].
- Add condition: Status = Requested. Click Done.

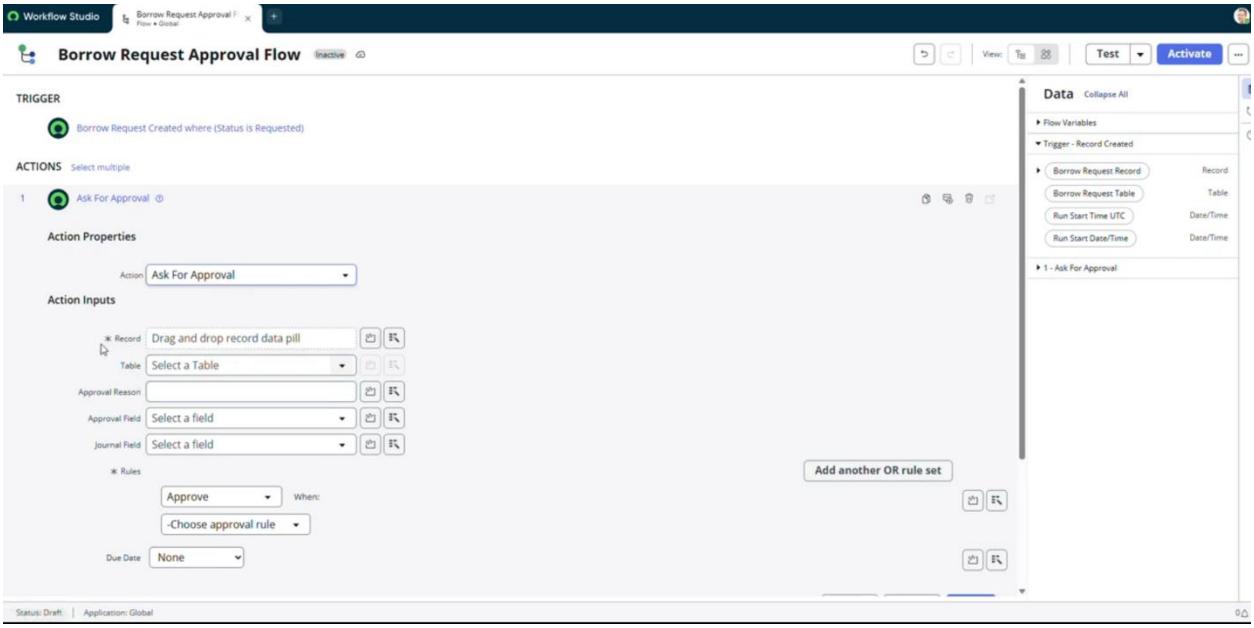


ADD ACTION – APPROVAL:

- Click on + Add Action.
- Search for Ask for Approval.
- Set Approvers = Role: librarian.
- Approval conditions:
  On Approve → continue flow.
  On Reject → set Status = Rejected.
- Click Done.

Add Action – Update Record (Issue Book):

- Click on + Add Action.
- Search for an Update Record.
- Choose Table = Book [u_book].
- Condition: Sys ID = Current.Book (the one from Borrow Request).
- Set Status = Issued.
- Click Done.

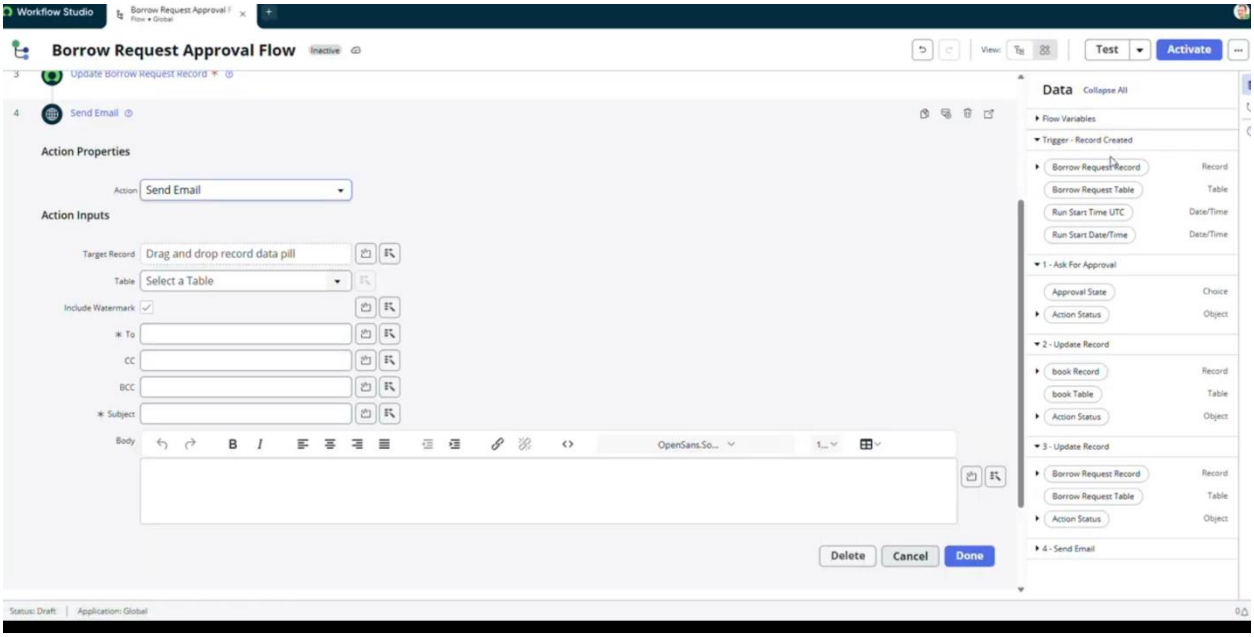


ADD ACTION – UPDATE BORROW REQUEST RECORD:

- Click on + Add Action.
- Search for an Update Record.
- Choose Table = Borrow Request [u_borrow_request].
- Condition: Sys ID = Current.
- Set Status = Approved.
- Click Done.

. **Add Action – Send Email (Notification**):

- Click on + Add Action.
- Search for Send Email.
- Set To = Requested By (Borrow Request → Requested By field).
- Subject: Your Borrow Request has been Approved.
- Body: The book [Book.Title] has been issued to you. Please collect it from the library.
- Click Done

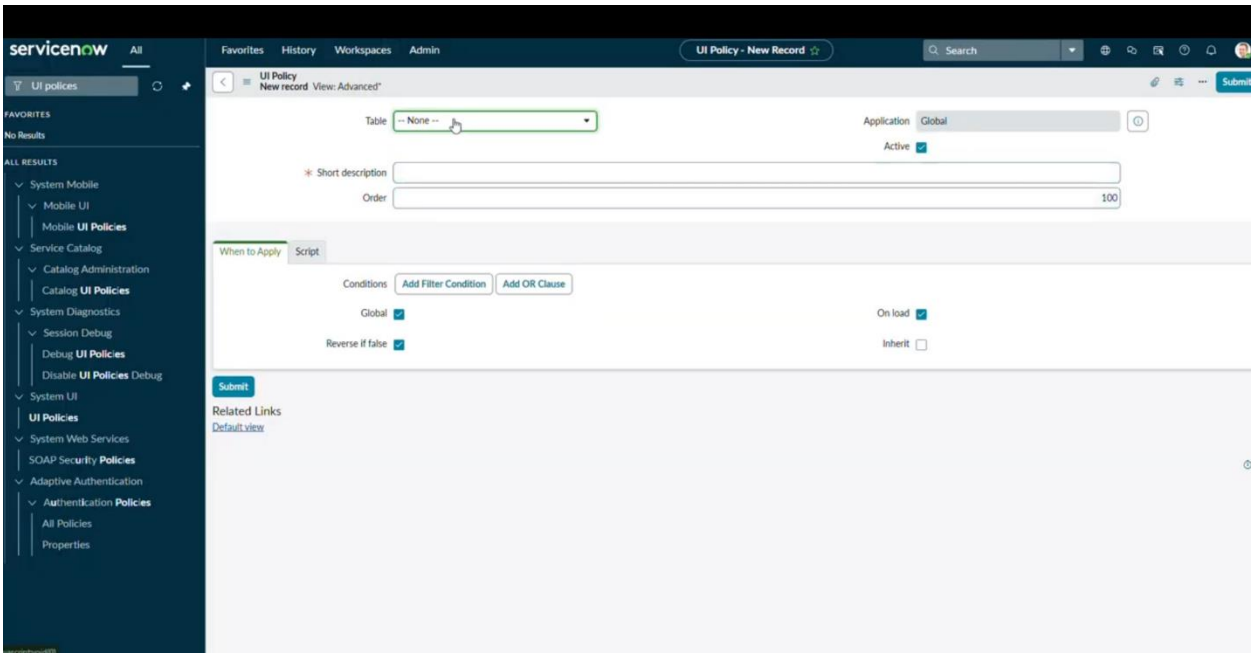


SAVE AND TEST FLOW:

- Click Save → Activate.
- Create a test Borrow Request with Status = Requested.
- Verify: Librarian gets approval → On approve, Book.Status = Issued, Borrow Request.Status = Approved, and student receives an email.

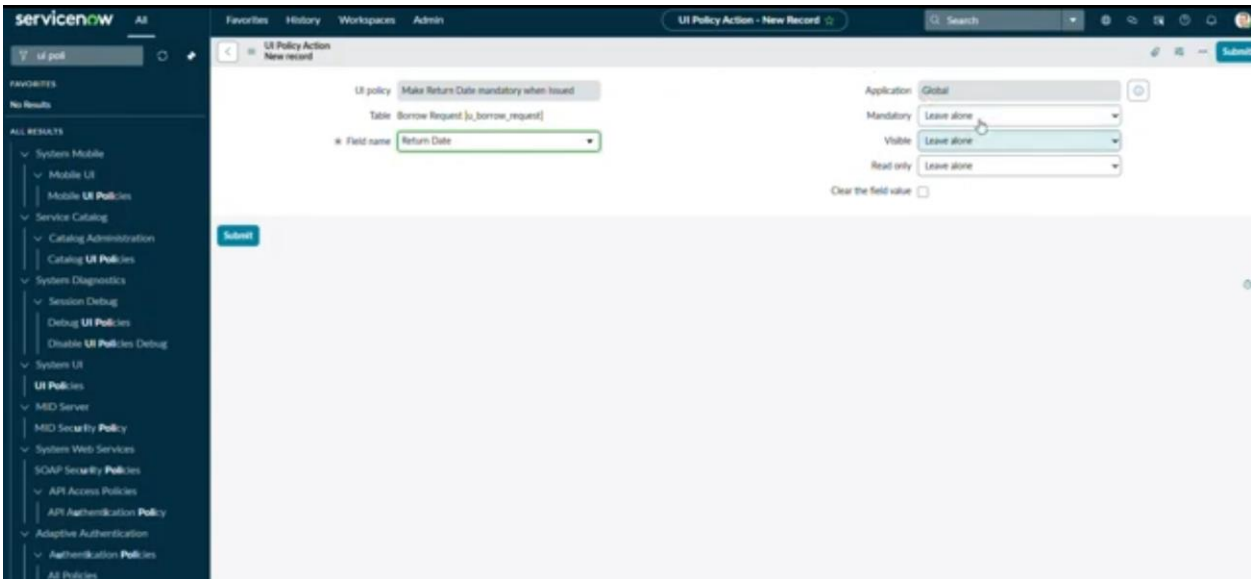
Create UI Policy:

1. Open ServiceNow.
2. Click on All → search for UI Policies.
3. Select System UI → UI Policies.
4. Click on New.
5. Enter Table = Borrow Request [u_borrow_request].
6. Enter Short Description = Make Return Date mandatory when Issued.
7. Submit the record.
8. Reopen the created UI Policy record.
9. In the Conditions section, set: Status = Issued.
10. In the UI Policy Actions related list, click New.
11. Select Field = Return Date.
12. Set Mandatory = True.
13. Set Visible = True.
14. Submit the UI Policy Action.



Creation Of ACL:

1. Open ServiceNow.
2. Click on All → search for Access Control.
3. Select System Security → Access Control (ACL).
4. Click New.
5. Select Table = Book [u_book].
6. Choose Operation = read.
7. Set Requires role = student, librarian.
8. Submit.
9. Again, click New.
10. Select Table = Book [u_book].
11. Choose Operation = write.
12. Set Requires role = librarian.
13. Submit.
14. Again, click New.
15. Select Table = Book [u_book].
16. Choose Operation = delete.
17. Set Requires role = librarian.
18. Submit.
19. Result: Students can only read books, librarians can read, write, and delete.



Creation of Reports:

1. Open ServiceNow.
2. Click on All → search for Reports.
3. Select Reports → View / Run.
4. Click on New.
5. Enter Name = Most Borrowed Books.
6. Select Source Type = Table.
7. Select Table = Borrow Request [u_borrow_request].
8. Choose Type = Bar Chart (or Pie Chart as needed).
9. In the Group by field, select Book.
10. In the Aggregate field, select Count.
11. Add a filter: Status = Approved (to count only issued books).
12. Click Run to preview the report.
13. Adjust chart options (Title, Labels, Legend) as required.
14. Click Save. To publish,
15. click Sharing → Add roles (student, librarian).

Conclusion:

This project demonstrates how workflow automation in ServiceNow can transform a simple manual process into a structured, efficient system. Instead of relying on human memory and manual tracking, the system enforces rules, tracks data accurately, and ensures accountability.

If you present this right, it shows you understand real ServiceNow concepts—not just theory but actual implementation. That’s what matters.